

Polynomial Projectors and Compressed Surface Assembly

Two certified decomposition kernels for the
ProveIt unknot-recognition program

Research continuation prepared for Vladimir Reshetnikov
with ChatGPT; proofs, implementation, and reproducible checks

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Abstract

We develop two exact improvements to the decomposition machinery in `Topology/UnknotRecognition`. The algebraic improvement augments the existing scalar Fitting splitter with polynomial idempotents obtained from Berlekamp’s Frobenius fixed algebra. It can expose direct summands when a chosen chain endomorphism is invertible and its stable kernel is zero. All matching, homological, and recovered quantum types are preserved, and every transformed differential attachment is checked. An explicit connected family of two-term graded complexes has scalar endomorphism algebra $\mathbb{F}_{2^r} \times \mathbb{F}_{2^r}$. On a uniform candidate, the old zero-eigenvalue test succeeds with probability $2(2^r - 1)/2^{2r}$, whereas the polynomial-projector test succeeds with probability at least $1 - r/2^r$. This is a separation between candidate tests, not a claim about all search policies or about realizability by knot prefixes.

The geometric improvement assembles binary dihedral surface-cover patches across verified full-boundary seams. A spanning-forest gauge reduces the assembly to one dihedral monodromy problem per connected base component. Component counts, orientability, genus, residual boundary lifts, and marked sheet queries are computed in bit-polynomial time without enumerating the sheets. Closed outputs and nonorientable seams are allowed. An annulus example shows that identical local topology can assemble into two Klein bottles or one torus solely because the seam shift changes.

The package contains an optional production primary-splitting backend, a standalone geometric assembly API, independent checks, raw measurements, and a pinned integration patch. Neither result supplies the missing global bound on recognition states or hierarchy repairs. We give precise conditional cost statements and a research agenda that separates these remaining obligations from the operations now implemented.

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1 Results, scope, and relation to the existing program

The purpose of this continuation is to improve operations that can actually occur inside a complete recognition strategy, while keeping their cost and logical role explicit. The maintained code already combines structural certificates, invariant filters, diagram simplification, and an exact Khovanov fallback. It also contains experimental compressed algebra and surface-cover infrastructure. Two gaps in that infrastructure are particularly concrete: detecting scalar direct summands beyond the stable kernel of a candidate endomorphism, and composing multiple compressed cover presentations without discarding their attachments.

The main contributions of this package are as follows.

1. An implemented, certificate-checked extension of the scalar splitter. The new recognition option is `backend="primary"`; the exact-rank CLI option is `khovanov--primary`. Both preserve the standard default policy.
2. A fully specified family demonstrating why a polynomial projector can succeed when a Fitting test fails, together with exact success probabilities for uniform endomorphism sampling. This also distinguishes a weakness of a particular test from a weakness of the entire search strategy.
3. A checked geometric input model for assembling surface covers, a proof that a spanning-forest reduction preserves all monodromy, and a binary algorithm for the resulting topology and attachment queries.
4. A precise account of which remaining parameters would have to be bounded to obtain $n^{O(\log n)}$ recognition, supported by reproducible positive and negative experiments.

Berlekamp’s fixed-space construction, including its treatment of repeated irreducible factors, is classical [2]. Covering-space monodromy is also classical [3]. The new work here is their checked application to these particular repository interfaces, the explicit comparison family, and the implementation and validation of the resulting operations. We make no claim that polynomial idempotents or spanning-tree gauges originate here.

Object	What is established	Remaining limitation
Typed scalar complex	Exact polynomial-projector splitting of a chosen endomorphism	Candidate selection and live matrix dimensions remain uncontrolled globally
Surface-cover assembly	Exact topology and marked-sheet queries for checked complete seams	Extraction from arbitrary normal cuts and embedded 3-manifold provenance are not supplied
Complete recognizer	An additional exact optional backend	No general quasipolynomial running-time theorem
Measurements	Reproducible bounded experiments and independent comparisons	They are not worst-case or average-case complexity proofs

All statements about the baseline refer to the commit shown on the title page, rather than a moving branch. The source inspected includes `scalar_split.py`, `barcode_scan.py`, `surface_cover.py`, and the corresponding maintained synthesis sections [1]. Prior incoming reports already contain additional Jones and compressed-word work. This article does not present those results as new.

2 The complexity target and the available geometric framework

Throughout, n denotes the number of crossings in an explicit validated knot diagram. Binary surface multiplicities and explicit matrix dimensions are separate parameters. We use “quasipolynomial” for $\exp((\log(n+2))^{O(1)})$, and reserve the sharper expression $n^{O(\log n)} = 2^{O((\log(n+2))^2)}$ for that particular target.

This distinction matters even in the motivational literature. The March 2021 Oxford handout states the announced running time $2^{O((\log n)^3)}$, and describes a route through hierarchies of length $O((\log n)^2)$ with polynomially bounded surface complexity [7, slides 7 and 25]. The sharper target $n^{O(\log n)}$ would improve that displayed estimate. We consider both the general quasipolynomial goal and this stronger possibility, without identifying their exponents.

Lackenby’s July 2026 preprint gives a hierarchy algorithm and bounds its iteration count by $L(g+1)^L$, where L is maximal hierarchy length and g is maximal pattern complexity. Its Section 9 explicitly leaves the cost of some operations insufficiently specified for a running-time estimate. Its Theorem 14.4 gives polynomial-time compressed cutting in the number of uniform-type handles and the logarithm of extended normal weight, including parallelity-bundle base topology and vertical attachments [8, Proposition 9.1 and Theorem 14.4].

The Agol–Hass–Thurston orbit machinery provides an established general framework for doing normal-surface topology without expanding all normal discs [4]. The seam algorithm below is a restricted, particularly explicit arithmetic kernel; it is not a replacement for that general theory. Similarly, recent surface-curve algorithms compute geometric intersection numbers and isotopy of essential normal 1-manifolds in time polynomial in surface triangulation size and logarithmic curve weights [9, Theorems 1.1–1.2]. These provide natural neighboring interfaces for future attachment transport.

For the algebraic branch, Khovanov homology detects the unknot [5]. The maintained code uses its characteristic-two rank formulation and existing exact/capped continuation machinery. A scalar chain isomorphism preserves all additive continuations, so our new splitting step can enter that machinery without introducing an additional knot-theoretic recognition hypothesis. The issue is the cost of constructing and simplifying the complex, rather than soundness of an accepted scalar decomposition.

The two new kernels should therefore be assessed on two scales. Locally, their correctness and bit complexity are fully specified. Globally, their usefulness depends on whether a recognition algorithm can discover their input presentations economically and keep their dimensions and number of calls under control. Section 12 formulates that separation as explicit inequalities.

3 Typed scalar endomorphisms of a scanning complex

3.1 The input category and the permitted changes of basis

Let C be a finite chain complex in the additive characteristic-two cobordism category used by the scanner. Each object has a frontier matching m , a homological degree h , and a recovered relative quantum shift q . Write its scalar type as $\tau = (m, h, q)$, and let $V_\tau = \mathbb{F}_2^{d_\tau}$ be the multiplicity space of objects of that type. Put

$$D = \sum_{\tau} d_{\tau}, \quad V = \sum_{\tau} d_{\tau}^2.$$

Here D is an explicit live object count, and V is the number of scalar matrix variables. Neither is automatically small as a function of n .

A scalar typed endomorphism is a tuple $Q = (Q_\tau)_\tau$ with $Q_\tau \in \text{Mat}_{d_\tau}(\mathbb{F}_2)$ satisfying

$$Qd = dQ. \quad (1)$$

The equality is tested coefficient by coefficient in each relevant morphism space. Distinct morphism coefficients must not be collapsed to their support size or to one scalar. In particular, a coefficient encoded by the integer one need not be the identity between different matchings.

When all entries of an invertible scalar matrix stay within a type, its change of basis is a legitimate isomorphism in the original category. Recovering q first prevents a scalar isomorphism from mixing different quantum shifts. The existing implementation uses the relation

$$q_v - q_u = p - c_{uv} + 2\delta_{uv}$$

for a nonzero homogeneous entry from u to v , where the frontier has $2p$ endpoints, c_{uv} is its overlay-circle count, and δ_{uv} is the dot degree. All relevant cycles are checked for consistency. We preserve this existing restriction [1].

3.2 Why whole-component verification is necessary

We work on a whole connected component of the undirected differential graph. A projector on a convenient internal block need not commute with maps to other blocks. If those attachments are omitted, splitting can change later homology even when the internal matrix calculation is correct.

Lemma 3.1 (Typed projector splitting). *Let E be a typed scalar endomorphism of C satisfying $E^2 = E$ and $Ed = dE$. Then*

$$C \cong \ker E \oplus \text{im } E$$

by a typed scalar chain isomorphism. Every additive continuation functor preserves this direct sum and its chain isomorphism.

Proof. In each multiplicity space, $v = (v + Ev) + Ev$, the first term lies in $\ker E$, and the second lies in $\text{im } E$. Their intersection is zero because E acts as zero on its kernel and as identity on its image. The commutation relation makes both subspaces invariant under the differential. Choose bases for the two parts within each type and concatenate them. The resulting change of basis is invertible, and its conjugated differential has no cross terms. An additive functor sends the change of basis and its inverse to mutually inverse maps and preserves the displayed direct-sum decomposition. $\square$

The implementation checks exactly the content needed for this argument: types, invertibility, the coefficientwise commutator, the transformed entries, and absence of every cross term. It also records the original and transformed rows when witness recording is enabled. A cache hit is not an exemption from commutator and basis-change verification.

3.3 The existing Fitting test

For a candidate Q , the old method chooses a power of two N at least the largest type-block dimension. Then $\ker Q^N$ and $\text{im } Q^N$ are stable and form a direct sum. If both are nonzero, they define a valid decomposition. The method is inexpensive for a useful nilpotent part.

However, an invertible Q always has zero stable kernel. This says nothing about whether Q has two different nonzero primary factors. It also says nothing about whether the full commutant $\text{End}(C)$ contains an idempotent. The new operation addresses the first limitation for the same candidate; it does not claim to solve the second problem for every complex.

4 Polynomial projectors from the Frobenius fixed algebra

Let $\mu_Q(t)$ be the minimal polynomial of the direct sum of the blocks Q_τ , and let $d = \deg \mu_Q \leq D$. Evaluation gives an injective algebra map

$$A_Q = \mathbb{F}_2[t]/(\mu_Q) \longrightarrow \prod_{\tau} \text{End}(V_{\tau}), \quad [p] \longmapsto p(Q).$$

Injectivity is precisely the minimal-polynomial property. Every image commutes with the differential because Q does.

Theorem 4.1 (Berlekamp fixed algebra, used as a splitting test). *Write $\mu_Q = \prod_{j=1}^s f_j^{e_j}$ with distinct monic irreducible f_j . Then*

$$\mathcal{B}_Q = \{[p] \in A_Q : [p]^2 = [p]\}$$

is an s -dimensional vector space over $\mathbb{F}_2$. It contains a nonconstant class exactly when $s \geq 2$. Every class other than 0 and 1 gives a nonzero, nonidentity typed projector $p(Q)$, hence a chain splitting.

Proof. The Chinese remainder theorem gives $A_Q \cong \prod_j \mathbb{F}_2[t]/(f_j^{e_j})$. Each factor is local. An idempotent e in a local ring is zero or one: if e is a unit then $e^2 = e$ implies $e = 1$; otherwise $1 - e$ is a unit and $e(1 - e) = 0$ implies $e = 0$. Thus the idempotents correspond exactly to binary vectors in $\mathbb{F}_2^s$. In a commutative ring of characteristic two, $(u + v)^2 = u^2 + v^2$, so these idempotents are indeed closed under addition and scalar multiplication. Evaluation is injective, preserving zero, identity, multiplication, and types. Finally apply Lemma 3.1. $\square$

The theorem, including repeated powers $e_j > 1$, is part of the classical Berlekamp method [2, pp. 1854–1858]. We give the proof to make clear why no squarefree assumption and no algebraic extension of the coefficient field is hidden in our use of it.

4.1 A factorization-free implementation

In the basis $1, t, \dots, t^{d-1}$ of A_Q , the map

$$\Phi : [p] \longmapsto [p^2 + p]$$

is binary linear. Its j th column is the coefficient vector of $t^{2j} + t^j$ modulo μ_Q . We compute its nullspace by packed binary elimination. The nullspace contains 1; if its dimension is at least two, at least one basis member is different from 0 and 1 and supplies a projector. We do not need to factor μ_Q first.

The minimal polynomial is obtained from the first linear dependence in

$$I, Q, Q^2, \dots, Q^D.$$

For each power, the blocks are flattened into one binary vector of length V . Elimination retains both a pivot vector and its polynomial expression. The first dependence has leading coefficient one and is the minimal polynomial: a lower-degree annihilator would be an earlier dependence. Cayley–Hamilton guarantees a dependence by degree D .

The discovery path evaluates $p(Q)$ using its retained powers. A separate positive-certificate checker evaluates the displayed polynomial by Horner’s rule, verifies an annihilating polynomial, checks the fixed-polynomial identity, and squares the proposed matrix. This checker need not prove that the supplied annihilator is minimal. For a positive certificate, actual idempotence, nontriviality, and chain compatibility suffice.

Remark 4.2 (What a negative result means). If $\dim \mathcal{B}_Q = 1$, then μ_Q is a power of one irreducible and no nontrivial polynomial in this Q is idempotent. The ambient complex may still split. For example, $Q = I$ on two identical direct summands has $\mu_Q = t + 1$, yet projection onto one summand is a perfectly valid non-polynomial idempotent. A failed candidate therefore retains the exact component and continues the bounded search.

4.2 A worked six-dimensional candidate

Take

$$f = t^3 + t + 1, \quad g = t^3 + t^2 + 1, \quad \mu = fg = t^6 + t^5 + t^4 + t^3 + t^2 + t + 1.$$

Both cubics are irreducible over $\mathbb{F}_2$: neither has a root in $\mathbb{F}_2$. The polynomial

$$p = t^4 + t^2 + t$$

has residues 0 modulo f and 1 modulo g . Consequently $p^2 = p \pmod{\mu}$. On $Q = C_f \oplus C_g$, where C_f, C_g are companion matrices, $p(Q)$ is projection onto the second three-dimensional summand. The candidate Q is invertible because $f(0) = g(0) = 1$, so its ordinary stable-kernel split is trivial. Conjugating both matrices by any invertible binary change of basis preserves these conclusions while hiding the coordinate decomposition.

4.3 Elementary bit bounds

Proposition 4.3 (Cost of a chosen candidate). *For typed binary blocks of total dimension D , the elementary implementation finds a polynomial projector or proves that none exists in $\mathbb{F}_2[Q]$ in $O(D^4)$ bit operations and $O(D^3)$ bits of auxiliary storage. Positive projector verification is polynomial in D . These bounds exclude the initial construction of the scalar endomorphism space and the differential basis change.*

Proof. One block-matrix product costs at most $O(\sum_{\tau} d_{\tau}^3) \leq O(D^3)$ elementary binary operations. There are at most D products to generate powers. Their packed vectors have length $V \leq D^2$, and eliminating at most $D + 1$ such vectors costs $O(D^2V) \subseteq O(D^4)$. Retaining D vectors of length V costs $O(D^3)$ bits. The Frobenius matrix has dimension at most D , and its construction and nullspace are within the same bound by ordinary polynomial reduction and elimination. Polynomial evaluation and squaring also fit. Packed Python integer operations may improve constants but are not counted as constant-time operations on unbounded integers. $\square$

If there are E binary commutation equations in V variables, ordinary elimination has the conservative bound $O(EV^2 + V^3)$; equation assembly and morphism transformation must also be charged. The production caps on objects, variables, candidates, and splits remain important for exactly this reason. A spectral test should not be described as polynomial in the logarithm of an unexpanded matrix dimension.

4.4 Simultaneous primary refinement

The full Berlekamp space also describes a finite refinement procedure. Starting from the identity idempotent, refine a current idempotent e by the pair ep_j and $e(1 + p_j)$ for each fixed-space basis member p_j , discarding zero pieces. These products remain orthogonal idempotents. In the Chinese remainder coordinates, the procedure separates the s distinct coordinates, so at most s nonzero final pieces remain. This is the classical simultaneous factor-separation mechanism [2].

The current production path takes one projector and lets its existing component recursion continue. It does not implement a special batched all-primary basis change. Avoiding repeated

construction of commutant systems after a large spectral split is a concrete future optimization, with a well-defined algebraic target.

5 A connected comparison family and exact success probabilities

This section constructs valid scanner-category complexes rather than assuming that arbitrary binary matrices arise during a knot scan. We will explicitly leave the latter realization question open.

5.1 The family

Let $r \geq 3$, choose distinct irreducible monic polynomials $f, g \in \mathbb{F}_2[t]$ of degree r , and put $q = 2^r$. On one matching with two arcs, choose its two dot variables x, y in

$$R = \mathbb{F}_2[x, y]/(x^2, y^2).$$

Set $A_0 = \text{diag}(C_f, C_g)$ and let

$$S = \begin{pmatrix} I & E \\ 0 & I \end{pmatrix}, \quad A = SA_0S^{-1} = \begin{pmatrix} C_f & C_fE + EC_g \\ 0 & C_g \end{pmatrix},$$

where E is any nonzero rank-one $r \times r$ matrix, for example $E = e_1e_1^\top$. Define the two-term complex

$$C_{f,g}: R^{2r} \xrightarrow{xI+yA} R^{2r}. \quad (2)$$

There are no further differentials, so $d^2 = 0$ automatically. Every nonzero coefficient has dot degree one, and the two homological layers have a consistent difference of quantum shift two. Hence the scalar-type restriction used by the production splitter permits the entire matrix in each layer.

Lemma 5.1. *The differential graph of (2) is connected, and its typed scalar chain-endomorphism algebra is isomorphic to $\mathbb{F}_q \times \mathbb{F}_q$.*

Proof. The coefficient of x in a chain-commutation equation forces the source and target scalar matrices to agree. The coefficient of y then says that this common matrix commutes with A . The companion module of an irreducible polynomial is a simple cyclic $\mathbb{F}_2[t]$ -module with endomorphism field $\mathbb{F}_q$. There are no nonzero homomorphisms between the two modules because their annihilators are coprime. Thus the centralizer of A_0 , and therefore of A , is the displayed product of fields.

For connectivity, the xI edges join the corresponding source and target vertices. Within either companion block, the shift entries of the companion matrix connect all indices through these edges. The off-diagonal block $C_fE + EC_g$ is nonzero: if it were zero, E would be a nonzero module map between the two nonisomorphic simple modules. One of its y edges therefore joins the two already connected blocks. No cancellation with x can remove this edge, because x and y are independent coefficients. $\square$

The complex is minimal in the sense that its differential coefficients lie in the radical (x, y) . Its hidden scalar splitting is not merely deletion of a scalar unit. This makes it a relevant stress model for the actual category, although no assertion is made that it is a reachable reduced prefix of a classical knot.

5.2 Fitting and primary success on a uniform candidate

Under the isomorphism of Lemma 5.1, a uniformly sampled scalar endomorphism is a pair $(a, b) \in \mathbb{F}_q^2$. Sampling is uniform when independent fair binary coefficients are used for a complete $\mathbb{F}_2$ -basis of the commutant. This is a mathematical comparison model. The production policy first tries a bounded list of basis vectors and then a fixed-seed list of combinations, so its measured behavior must be reported separately.

Theorem 5.2 (Exact candidate-test separation). *For a uniform candidate on $C_{f,g}$, the ordinary stable-kernel test has success probability*

$$P_{\text{Fit}} = \frac{2(q-1)}{q^2}. \quad (3)$$

Let $I_2(d)$ be the number of monic irreducible polynomials of degree d over $\mathbb{F}_2$. The polynomial-projector test has success probability

$$P_{\text{Prim}} = 1 - \frac{1}{q^2} \sum_{d|r} d^2 I_2(d) \geq 1 - \frac{r}{q}. \quad (4)$$

Thus repeated independent Fitting trials need an expected $q^2/(2(q-1))$ candidates. Repeated independent primary trials need at most $(1 - r/q)^{-1}$ candidates.

Proof. Multiplication by a nonzero field element is invertible, and multiplication by zero is zero. A nontrivial stable kernel/image decomposition therefore occurs exactly when one of a, b is zero and the other is nonzero, giving (3).

The minimal polynomial of the direct-sum action is the least common multiple of the minimal polynomials of a and b . Each individual minimal polynomial is irreducible. Consequently the Berlekamp dimension is one exactly when a and b have the same minimal polynomial. An irreducible of degree d has all its d distinct roots in $\mathbb{F}_q$ exactly when $d \mid r$. Its contribution to the number of ordered pairs with the same minimal polynomial is d^2 . This proves the equality in (4). For each a , its Frobenius orbit contains at most r possible b , giving at most rq failing pairs and the inequality. Expected trial counts follow from the geometric distribution. $\square$

The separation is exponential in r for the *number of independent candidate tests*. It is not automatically an exponential end-to-end speedup: constructing the commutant, the relative per-candidate costs, and the possibility that a deterministic basis vector already splits must all be counted. The elementary costs above are polynomial in r , so the uniform-sampling comparison still gives an exponential expected-cost separation for these two isolated test strategies once their common commutant has been supplied. Section 11 separately records the actual bounded policy.

5.3 A universal information limit of one candidate

The product-field family is favorable because its two field values usually have different minimal polynomials. If several summands receive Frobenius conjugate values, the same polynomial p cannot distinguish them: evaluation in $\mathbb{F}_2[Q]$ sees only primary factors, not an independently labeled copy of each summand. This is a structural limitation of every polynomial-in-one-endomorphism scheme, not a numerical failure. Additional candidates or noncommutative commutant information are required for those collisions.

Conversely, an accepted projector is useful even if it does not expose every indecomposable summand. Literal component sharing and interval normalization may benefit from a partial split. The new method therefore composes with the existing exact operations rather than replacing their correctness arguments.

6 A checked input model for assembling compressed surface covers

6.1 Pieces, canonical peripheral systems, and binary fibres

A piece is a connected compact surface F_v with nonempty boundary, given by its orientability, genus or crosscap number, and number b_v of boundary components. A canonical polygon schema supplies a free generating system for $\pi_1(F_v)$, an orientation bit for each generator, and based peripheral words for its boundary circles. An orientable genus- g_v piece has free rank $2g_v + b_v - 1$; a nonorientable piece with k_v crosscaps has rank $k_v + b_v - 1$. Handle generators have orientation bit zero, crosscap generators have bit one, and peripheral generators have bit zero.

In this section every piece has the same sheet set

$$X = \mathbb{Z}/W\mathbb{Z}, \quad W \geq 1,$$

with W encoded in binary. Each free generator acts as an affine permutation

$$g(x) = sx + a \pmod{W}, \quad s \in \{1, -1\}. \quad (5)$$

Since the fundamental group of a bordered surface is free, any assignment of these maps to its free generators defines a covering. The boundary monodromies $B_{v,j}$ are then *derived* from the canonical peripheral words. The caller cannot supply unrelated boundary maps together with a claimed genus and expect the kernel to certify a surface.

The covering of a piece can be disconnected. A connected component of the piece-adjacency graph will ultimately have an ordinary degree- W covering over its connected sewn base surface, possibly with many connected covering components. Different adjacency components can be processed independently. The current API chooses one common W for the whole request; varying W across independent requests is immediate. Unequal sheet counts across a seam are outside the interface.

6.2 Seams and their exact compatibility condition

A seam e uses two distinct, previously unused boundary components (u, a) and (v, b) . The pieces u, v may be equal. The seam specifies:

- a base boundary homeomorphism of degree $\varepsilon_e \in \{1, -1\}$;
- a fibre bijection $A_e(x) = s_e x + a_e \pmod{W}$.

Boundary circles have fixed based parametrizations; the base map is the canonical based map of degree ε_e , or a map in the same based isotopy class. The seam joins the *entire* inverse image of the first boundary to the entire inverse image of the second. Every lifted circle participates. Its necessary and sufficient compatibility check is

$$A_e B_{u,a} = B_{v,b}^{\varepsilon_e} A_e \quad \text{as permutations of } X. \quad (6)$$

Lemma 6.1 (Realization of a seam). *Condition (6) is equivalent to existence of an isomorphism of these complete boundary covers lying above the prescribed degree- ε_e base map and inducing A_e on the reference fibre. Compatible seams assemble the piece coverings into an unbranched covering of the sewn surface, including when that surface is closed.*

Proof. Represent a boundary covering by $X \times \mathbb{R}$ with $(x, t+1) \sim (Bx, t)$. Following a once-around base loop before or after the proposed fibre map must agree, which gives (6). Conversely

the map $(x, t) \mapsto (A_e x, \varepsilon_e t)$ respects these identifications exactly when (6) holds. It therefore descends to the required circle-cover homeomorphism and induces A_e at the reference fibre $t = 0$. Gluing collars along these homeomorphisms gives covering charts across the seams; away from them the original covering charts remain. Since each boundary is used once, the quotient base and covering are surfaces and no branching is introduced. $\square$

Two formal affine pairs need not be distinct permutations when W is small. The precise equality test is

$$(s, a) = (t, b) \text{ on } X \iff s - t \equiv 0 \pmod{W} \text{ and } a - b \equiv 0 \pmod{W}. \quad (7)$$

Necessity follows by evaluating at zero and one; sufficiency follows by subtraction. For $W = 1$ every map is identical, and for $W = 2$ the signs 1 and -1 coincide as permutations. The code checks (7) rather than insisting on equality of the written signs. Otherwise it would reject valid one- and two-sheet gluings.

6.3 Orientation is additional data

The sign s_e of a fibre permutation is not a surface-orientation character. These are independent quantities. Choose a local surface orientation at each piece basepoint and fixed whiskers to its boundary reference points. Orient each boundary using that transported local orientation. In this convention, traversing a seam changes the transported surface orientation by

$$\sigma_e = [\varepsilon_e = +1] \in \mathbb{F}_2. \quad (8)$$

An orientation-reversing identification of induced boundary directions extends the same surface orientation over the seam, while an orientation-preserving one requires its reversal. This explains the indicator in (8).

For a nonorientable piece the canonical polygon schema still fixes the whiskers and the local orientation choices. Orientation character on a based loop is well defined even though a global orientation does not exist. The implementation retains the character through every gauge change.

7 Spanning-forest reduction and attachment-preserving monodromy

Let Γ be the multigraph whose vertices are pieces and whose edges are seams. Loops and parallel edges are allowed. Choose one root and a spanning tree in each connected component. For a piece v , let P_v be the fibre transport from its root along the chosen tree path, and let $t_v \in \mathbb{F}_2$ be the corresponding orientation-transport bit. Set $P_r = \text{id}$ and $t_r = 0$. For a forward tree edge $u \rightarrow v$,

$$P_v = A_e P_u, \quad t_v = t_u + \sigma_e.$$

For a reversed tree traversal use A_e^{-1} ; the same σ_e applies. All P_v remain affine maps with sign ± 1 , irrespective of the path's length or the sheet count.

Transporting everything to the root yields the following data:

$$\hat{g}_{v,i} = P_v^{-1} g_{v,i} P_v, \quad \hat{\omega}_{v,i} = \omega_{v,i}, \quad (9)$$

$$H_e = P_v^{-1} A_e P_u, \quad \eta_e = t_u + \sigma_e + t_v \quad (e \text{ non-tree}), \quad (10)$$

$$\hat{B}_{v,j} = P_v^{-1} B_{v,j} P_v, \quad ((v, j) \text{ unsewn}). \quad (11)$$

Products mean composition, with the rightmost map acting first.

Theorem 7.1 (Monodromy of the assembled cover). *For each connected component of Γ , the action on the root fibre generated by (9) and (10), with their displayed orientation bits, is the monodromy and orientation character of the assembled surface covering. Its orbits are exactly the covering components. Formula (11) gives the residual peripheral monodromies. A point in piece v with local sheet x belongs to the root orbit of $P_v^{-1}(x)$.*

Proof. Consider a loop in the sewn surface. Subdivide it whenever it crosses a seam. Within a piece, replace each segment by the fixed whiskers and a word in the piece generators. Insert tree paths from and back to the root between these segments. The tree transports cancel in consecutive products. The remaining root loops are generated by the conjugated piece loops and non-tree seam loops, with precisely the fibre actions displayed above. Conversely every listed generator is represented by such a based loop in the sewn surface. This proves equality of the monodromy groups, not just one inclusion.

Orientation bits add along paths. A conjugated local loop contributes $t_v + \omega_{v,i} + t_v = \omega_{v,i}$. A non-tree loop contributes $t_u + \sigma_e + t_v$. The peripheral and point formulas are the same root-path calculation. The standard monodromy correspondence identifies connected covering components with orbits of the root fibre. $\square$

This proof can be viewed as an explicit groupoid form of the classical covering-space action construction [3, Section 1.3]. It avoids rewriting the sewn surface into a new canonical polygon, a step that could otherwise discard useful labels or introduce large words.

7.1 Closed bases are handled by actual gluing

Once all boundaries are sewn, the base may be closed and its fundamental group is generally no longer free. We do not assert that an arbitrary list of affine maps defines a closed-surface covering. Instead, Lemma 6.1 already constructs the covering. The relations of the closed base are therefore automatically satisfied by the assembled action. This is the reason the new kernel can safely extend the older bordered-cover interface to closed outputs.

For each connected base component F , Euler characteristic is

$$\chi(F) = \sum_{v \in \Gamma_F} \chi(F_v). \quad (12)$$

Gluing circles changes no Euler characteristic. If C is a connected covering component of degree d_C , lifting a cell decomposition gives

$$\chi(C) = d_C \chi(F). \quad (13)$$

Boundary counts and orientability then determine genus. These formulas count actual covering degrees and topological characteristics; neither multiplicities nor Euler characteristics are reduced modulo two.

7.2 A replayable gauge certificate

The assembly stores a record for every piece: its root, parent, tree seam, traversal direction, affine transport, and orientation bit. An independent checker validates the original seam equations, verifies every parent edge and transport recurrence, rejects cycles, and checks that all seams stay within the claimed forest components. Parent chains are settled once rather than re-walked independently from each vertex.

This checker certifies the gauge reduction. It deliberately does not accept an arbitrary serialized genus or component count as a verified topological answer. The prepared index is reconstructed

from the validated input; the arithmetic classifier then computes its summary. Independent expanded-sheet audits provide a separate implementation check on small inputs.

Changing the chosen root or tree changes coordinates but not the assembled cover. The verifier accepts a valid alternative tree certificate. The production index nevertheless retains its own normalized presentation and labels, so its equality keys are defined within that presentation rather than claiming to solve isomorphism of arbitrary surface decompositions.

8 Exact orbit, orientation, and marking arithmetic

The root generators still have the form $x \mapsto \pm x + a \pmod{W}$. We now recall and justify the arithmetic classification needed after assembly. This is the extension point at which the existing dihedral cover kernel is reused [1].

8.1 Translation-generated subgroup and component families

If every generator has positive sign, set

$$d = \gcd(W, a_1, \dots, a_k), \quad m = W/d.$$

The translation subgroup is generated by $T(x) = x + d$. Its orbits are the d residue classes modulo d , each of size m .

If reflections occur, choose one $R(x) = -x + a_0$. Define H as the subgroup generated by all displayed positive shifts and all products of two displayed reflections. Its translation shifts are generated by the positive shifts and the differences $a_i - a_0$ for the other reflections. Take their gcd with W as d and again let $m = W/d$. Products of two reflections give these differences, and conjugating a translation by a reflection negates it. Bezout's identity proves that these listed translations generate exactly $d\mathbb{Z}/W\mathbb{Z}$, so $H = \langle T \rangle$ and the full action is generated by H and R . For $W \geq 3$, H is the usual translation subgroup of that action. For $W \leq 2$, formal signs are auxiliary labels and H need not contain every group element that happens to act as a translation; the displayed generation statement remains valid.

The reflection acts on residue classes modulo d by $r \mapsto a_0 - r$. Let

$$f = \#\{r \pmod{d} : 2r \equiv a_0 \pmod{d}\}.$$

This number is zero, one, or two. There are $(d - f)/2$ paired-residue components of degree $2m$, together with f fixed-residue components of degree m . Thus at most three family records are required.

Proposition 8.1 (Uniform family bound). *For each connected sewn base component, the assembly has at most three component-family records, with multiplicities and covering degrees written in binary. All paired-residue components are isomorphic as unmarked covers of that fixed base. If two fixed-residue components occur, they are isomorphic exactly when m is odd.*

Proof. Only the last two statements need explanation. For nonfixed residues r, s , map the two translation orbits by

$$r + jd \mapsto s + jd, \quad a_0 - r + jd \mapsto a_0 - s + jd.$$

The resulting bijection commutes with T and R , hence with all root generators. For two fixed residues, an equivariant map must commute with the transitive translation action and is therefore a translation. Their residue difference is $d/2$ modulo d , so the translation has size $d/2 + dt$. Commutation with R requires $2t \equiv -1 \pmod{m}$, which is solvable precisely when m is odd. $\square$

The bound is per connected base component. A disconnected base with k components can require up to $3k$ records. An unmarked isomorphism type is also different from a particular component label: attachment queries must retain which residue class or pair is selected, even when two selected components happen to be isomorphic.

8.2 Orientation through a character-consistency calculation

Attach a bit $\omega(g)$ to each root generator. Write every translation generator as a shift a_j with bit β_j , and write the chosen reflection as (R, β_0) . A product of two reflections supplies the translation shift difference with the sum of their bits.

Use extended gcd to compute d as an integer combination of W and the translation shifts. Applying the same coefficients modulo two to the β_j gives a candidate bit τ for translation by d . The orientation character on the displayed translation words descends consistently to H exactly when

$$m\tau = 0 \pmod{2}, \quad \beta_j = (a_j/d)\tau \pmod{2} \quad \text{for every translation generator.} \quad (14)$$

If these fail, an orientation-reversing identity translation exists; it fixes every sheet, and every component is nonorientable.

Suppose (14) holds. A translation-only component is orientable. A paired-residue component is also orientable: a reflection changes residue classes and cannot stabilize a sheet; every stabilizing translation has bit zero. On a fixed residue class r , a reflection stabilizer is

$$T^{(2r-a_0)/d}R,$$

whose orientation bit is

$$\beta_0 + \frac{2r - a_0}{d}\tau \pmod{2}. \quad (15)$$

That component is orientable exactly when this bit is zero.

These criteria remain valid when formal affine signs act nonfaithfully for $W = 1, 2$. They inspect orientation bits of the actual stabilizers rather than assuming that a written reflection has a nontrivial permutation action. The repeated use of integer relations in (14) also explains why an arbitrary choice of a gcd representative cannot silently change the answer.

8.3 Residual boundary-cycle profiles

A remaining base boundary has a root affine monodromy B . If $B(x) = x + a$, then $d \mid a$. On one translation orbit of size m , it has

$$c = \gcd(m, a/d)$$

cycles, each of degree m/c . On a paired-residue component, there are two such translation orbits, so the multiplicity doubles.

If $B(x) = -x + a$, then on a paired-residue component every cycle has length two, with m cycles. On a fixed residue r , its action in the coordinate $x = r + jd$ is

$$j \mapsto -j + \frac{a - 2r}{d} \pmod{m}.$$

The number of fixed points is the number of solutions to $2j = (a - 2r)/d \pmod{m}$, hence at most two. All other points form two-cycles. The total number b_C of boundary circles is the sum of these cycle multiplicities over all unsewn base boundaries.

Combining with (13), an orientable component has genus

$$g_C = \frac{2 - b_C - d_C \chi(F)}{2};$$

a nonorientable component has crosscap number $2 - b_C - d_C \chi(F)$. Integrality and nonnegativity are checked. An invalid result is an arithmetic error, never an opportunity to guess a topological type.

8.4 Component keys and seam provenance

A local point (v, x) is first moved to the root sheet $z = P_v^{-1}(x)$. Its component key is its residue $z \pmod{d}$ in the translation-only case, or the unordered pair

$$\{z \pmod{d}, a_0 - z \pmod{d}\}$$

in the reflection case, together with the base-component label. A singleton pair denotes a fixed residue. The key is computed on demand and does not allocate one object for every component.

A boundary-lift key uses the port's local sheet x and its local boundary monodromy. For a translation on X that cycle is labeled by $x \pmod{\gcd(W, a)}$; for a reflection it is the one- or two-element set $\{x, a - x\}$. At a sewn port, the key additionally retains the seam identity and first transports a right-port sheet to the left port by A_e^{-1} . It then uses the left port's local boundary cycle. The global component key is computed separately using the root coordinate. Therefore the two sides of one lifted seam agree as an attachment, whereas unrelated circles with the same length remain distinguishable.

8.5 Ordered marks and equivariant transport

For a transitive group action, an equivariant map is uniquely determined by one point image when such a map exists. This gives efficient marked queries. First transport all marks to the root, retaining their piece labels. For translation-only components, a map taking anchor a to b is translation by $b - a$. For fixed-residue components, this translation is equivariant exactly when

$$2(b - a) = 0 \pmod{W}.$$

For paired-residue components, points in the translation orbit of a move by $b - a$, while points in its reflected orbit move by $a - b$.

The same root map extends over piece v as $P_v f P_v^{-1}$. This proves the local point-transport formula and shows why preserving piece labels in ordered-mark signatures is necessary. Repetition of a marked point and the order of marks are both meaningful. The sheet used to select a component is not an additional mark: a query with no marks returns its cover type, and a query with marks anchors its normalization at the first actual mark. The interface classifies ordered point-marked components over one fixed checked assembly; it does not solve arbitrary embedded-curve equivalence.

The prepared index retains its normalized piece and seam presentation. Adding another valid seam rebuilds from those retained data, not from the short topology summary. This immutable continuation operation keeps the original index usable if the new seam is invalid or a cooperative check interrupts its construction. Summary compression therefore does not erase information required for future assembly.

9 Geometric cost, examples, and limitations

Let K count the explicit piece generators, canonical peripheral-word letters, boundaries, seams, and piece records. Let B bound the bit lengths of W and the supplied integer shifts, increased by $O(\log(K + 2))$ for resulting Euler and multiplicity arithmetic. The integer shifts are normalized modulo W after being read; the cost of reading a very large supplied shift is part of the input cost.

Theorem 9.1 (Bit-polynomial surface assembly). *For a validated input of the preceding model, the assembled cover’s family summary, gauge certificate, and prepared attachment index can be constructed in $\text{poly}(K, B)$ bit time and space, without enumerating sheets or lifted boundary circles. An elementary conservative bound for the arithmetic construction is $O(KB^3 + K \log(K + 2))$ bit operations, with the input and output accounting included. A fixed number of point or port queries uses polynomial time in B after preparation; an ordered q -point query also charges its q entries and its returned presentation key.*

Proof. Canonical words have total length $O(K)$. Affine composition and inversion use signs, additions, and reductions on $O(B)$ -bit integers; no fibre permutation is materialized. Forest construction visits $O(K)$ vertices and edges. Conjugated local, seam, and boundary maps require $O(K)$ further affine operations. There are $O(K)$ gcd or extended-gcd updates. The conservative schoolbook bound of $O(B)$ Euclidean iterations, each with $O(B^2)$ arithmetic, gives $O(B^3)$ per update. Each connected base has at most three records, and every boundary contributes at most two cycle-length records per family. Genus and multiplicity computations involve integers of $O(B)$ bits. Sorting labels and output grouping fits the displayed bookkeeping bound. Point and port formulas use the already prepared gauges, divisors, and maps. $\square$

The cubic arithmetic estimate is deliberately elementary. Faster integer arithmetic and sharper gcd analysis can improve it, but are not needed for the key distinction: dependence on $\log W$, rather than on W . Serializing an entire defensive copy of the index or a signature containing the whole normalized presentation also costs its full output size. We do not label those operations constant time.

Repeated immutable extensions reconstruct an index of the current size. For M added seams, a conservative total is the sum of the M reconstruction costs. The implementation makes no amortized dynamic-update claim. A future incremental algorithm would need to retain gauge and orbit data carefully across component merges and new cycle generators.

9.1 Identical annulus pieces, different sewn topology

Example 9.2 (Seam shift changes both connectivity and orientability). Take one annulus with $W = 2m$ sheets and core monodromy $g(x) = x + 2$. Its covering consists of two degree- m annuli. Sew the two base boundaries with $\varepsilon = +1$ and map $A_b(x) = -x + b$. The boundary monodromies are g and g^{-1} , so $A_b g = g^{-1} A_b$, and the seam is valid for every b . The base becomes a Klein bottle.

The translation divisor is $d = 2$ and the translation orientation bit is zero. The seam contributes reflection character one.

Shift	Residue action	Sewn components	Total covering degree
$b = 0$	Both parity classes fixed	Two Klein bottles	$m + m = 2m$
$b = 1$	Parity classes exchanged	One torus	$2m$

For $b = 0$, each class has the orientation-reversing stabilizer in (15); with Euler characteristic zero and no boundary, its component is a Klein bottle. For $b = 1$, there is one paired-residue component and no orientation-reversing stabilizer. It is an orientable closed surface of Euler characteristic zero, hence a torus.

All local piece topology is identical in the two cases. Discarding b or replacing the piece by a list of unmarked annulus types would make a future seam operation unsound. The example remains valid for $W = 2$, and can be computed with $W = 2^{24000}$ by the same formulas.

These are abstract covering-space examples. In particular, we do not claim that the closed nonorientable outputs occur as embedded surfaces in an arbitrary knot exterior in S^3 . Their role is to prove necessity of the attachment data and to test the declared geometric model.

9.2 Capping a Mobius cover

A Mobius band has one generator a and boundary word $a^{\pm 2}$, depending on the chosen convention. Gluing a disk along the boundary requires that the square of its monodromy be the identity, since the disk's boundary monodromy is trivial. The seam check enforces this automatically.

Identity monodromy gives projective-plane components after capping. A nontrivial two-cycle with the Mobius generator's orientation character one has an orientable degree-two covering component; capping yields a sphere. For a reflection on an arbitrary finite cyclic sheet set, its fixed sheets and two-cycles give the corresponding mixture. This tests an actual closed surface relation arising from a cap, rather than an arbitrary action paired with an invented Euler characteristic.

9.3 Pants and long gauge chains

Sewing two pairs of pants along all three boundaries using compatible orientation-reversing base maps produces a closed orientable genus-two base. Compatible affine monodromies give covering components whose Euler characteristics are twice the negative covering degrees, so a connected degree- d_C component has genus $1 + d_C$. Tests include this family with nontrivial gauge maps.

A long chain of annuli gives a complementary check. It can have arbitrarily many pieces but a simple final base. The gauges compose all seam maps, including reversed tree traversals and large shifts. An explicit-sheet implementation would allocate vertices for every piece-sheet pair; the compressed kernel retains one affine gauge per piece. Such a chain tests scaling with K independently of scaling with B .

9.4 The boundary of the theorem

The kernel does not accept partial interval identifications, unequal fibre sizes across a seam, branching, arbitrary permutation monodromy, or unspecified normal-surface attachments. These extensions may be possible, but their correctness and cost require additional arguments. In particular, a general normal cut can produce interval-pairing data that has no single global signed-affine coordinate of this form.

Even a fully correct surface-topology summary does not certify a 3-ball, a solid torus, an essential boundary pattern, or a knot verdict. The gluing API therefore has its own command and returns geometric data only. The missing connection to complete recognition is a certified extractor from actual normal cuts or parallelity bundles, retaining the embedded attaching information that the hierarchy algorithm needs.

10 Production integration and resource contracts

10.1 An optional algebraic backend

The production change adds `fitting_primary=False` to the existing Fitting scanner and exact and capped wrappers. With the option enabled, an examined scalar endomorphism first receives the original stable kernel/image test. If that test fails and the candidate could still supply a new primary decomposition, the scanner invokes the polynomial-projector routine. Nilpotent candidates and literal identity operators are skipped: neither can have a nontrivial polynomial idempotent. The default policy, object limit, scalar-variable limit, candidate sequence, and split budget are unchanged.

The new recognition backend `primary` routes through the capped interval-decision wrapper. Its evidence names the selected policy and records primary candidates, primary splits, and maximum encountered minimal polynomial degree. Exact scanning reports `scalar-primary-interval-sharing`; capped decision reports `scalar-primary-interval-decision`. The policy is included in negative cache keys, so a previously cached failure of the original test cannot suppress the stronger search.

The recognizer’s existing early certificates and one-sided filters remain available. Consequently selecting the primary backend need not invoke any primary calculation: an earlier certificate may settle the diagram, or ordinary Fitting splitting may settle every candidate that the scalar policy examines. The raw-scan experiments below deliberately bypass early recognition exits so that this distinction is observable.

Option compatibility is checked before early certificates. Unsupported combinations with other scanner policies, nonstandard composition or reduction modes, and unrelated acceleration backends are rejected rather than silently changing the requested computation. The exact-rank CLI’s `--primary` option follows the existing Fitting command’s uncapped contract. For an explicit object or time budget, use the recognition CLI or the capped Python API.

10.2 Positive checks, incomplete search, and interruption

There are three different logical outcomes of primary discovery:

1. A verified nontrivial projector supplies an exact direct sum. The basis change is committed only after the transformed differential and cross terms have been checked.
2. A tested operator has no nontrivial polynomial projector. This says nothing about other operators in the commutant.
3. A search allowance is exhausted. This is a resource event, not a mathematical certificate of indecomposability or triviality.

Cooperative checks occur during polynomial construction, elimination, verification, and basis conversion. A failed or interrupted search preserves the current complex. The exact scanner can continue with it; the bounded recognizer preserves the established UNKNOWN semantics if its global allowance expires. A finite amount of optional simplification cannot turn an incomplete recognition attempt into an UNKNOWN verdict.

The positive polynomial verifier checks an annihilating polynomial, the polynomial congruence, the resulting matrix, and its idempotence and nontriviality. It does not establish the minimality of claimed metadata or certify a negative result. The full dense chain-witness checker is a separate layer: it verifies types, coefficientwise commutation, invertible basis matrices, both conjugation directions, and every differential attachment. This separation makes the trusted assertion of a successful split explicit.

10.3 The geometric API is a separate exact operation

The main interface is `GluedCoverIndex(raw)`. Its summary gives sewn base types and covering-component families. Other methods supply component keys, boundary and seam lift keys, ordered-mark signatures, and the unique point transport when a specified based covering isomorphism exists. The method `with_seams` creates a new checked index from the retained presentation and additional seams. A caller may provide a cooperative `check` function to preparation and extension.

The command `python-mfastunknot.surface_gluinput.json` exposes this operation without placing it on the knot-verdict path. Its optional `--port` query asks for a particular lifted attachment; its `--seconds` option bounds cooperative work. Large integers use the repository’s lossless integer codec, including hexadecimal encodings when needed. Summary dictionaries are defensive copies. Modifying a returned summary does not alter the prepared action or provide an accepted certificate for a later operation.

10.4 A carried-forward subprocess repair

The complete pinned baseline suite exposed one existing error in the normal surface subprocess wrapper: retrying `communicate` after a timeout could lose the still-unsent part of a large request. This package integrates the repair already supplied by the earlier incoming report `unknot_certified_primitives_20261008.zip`. It is included for a working integration and is not a new mathematical contribution here.

A dedicated communicator calls `communicate(payload)` exactly once. The supervising thread polls local and global cancellation, terminates and reaps the child when necessary, and joins the communicator before releasing the streams. The existing partial-input regression now passes. A narrowly scoped additional correction classifies a communicator’s decoding error from malformed worker bytes as an inconclusive worker failure. Its new regression covers both output streams, process reaping, and preservation of caller-originated exceptions. Optional Regina verdicts retain their external-engine evidence labels; this repair neither verifies Regina internally nor establishes a runtime bound for it.

11 Experiments and independent validation

The numerical evidence has three different purposes: checking exact identities, measuring a bounded decomposition search, and demonstrating the avoidance of explicit sheet expansion. No experiment below establishes a worst-case recognition bound or a distribution of natural knot instances. All raw samples, inputs, seeds, operation counts, and applicable witnesses are included in the package.

11.1 Environment and measurement protocol

The recorded runs used Python 3.12.14 on Linux 6.18.44, x86-64, with glibc 2.39. The two implementation kernels and their correctness tests use the Python standard library. The article’s plotting script additionally uses Matplotlib. No optional Regina installation was available during this validation.

For algebra, each experiment has seven shuffled paired rounds with ten calls per timed batch. An identical Fitting A/A control detects runtime variation. Every recursive-compression arm receives freshly constructed state; fixture construction is outside its timer. A natural-diagram scan includes the whole raw scan using a common preselected crossing order, but excludes

imports, diagram construction, and order selection. The table statistic is the median of the seven within-round Fitting/primary time ratios. It is not the ratio of two independent medians. For geometry, five measured shuffled rounds follow one excluded warmup. Small topology queries use batches of 100 compressed preparations and 100 identical controls per sample, versus one explicit expanded query. Binary scaling queries use batches recorded in the raw data. Full preparation and summary copying are inside the preparation timer; input construction and JSON serialization are outside. Serialized input, summary, signature, and gauge sizes are recorded separately. Timing entries are medians of the per-query batch times. No fitted empirical power law is used as a proof.

11.2 Natural-knot scans: a negative result that constrains the claim

Table 1: Complete raw-scan paired median time ratios. Values above one favor the primary policy; its kernel was dormant in every case.

Diagram	n	Fitting/primary	Fitting/control	Primary calls	Primary splits
Conway	11	0.995	0.994	0	0
Kinoshita–Terasaka	11	0.928	1.004	0	0
Hard unknot	8	1.001	0.995	0	0
Stress braid	36	0.962	0.977	0	0
$T(3, 5)$	10	1.067	1.012	0	0

Every one of these five scans examined *zero* primary candidates and performed *zero* primary splits. The stronger kernel was dormant under the retained caps and the earlier candidate outcomes. Thus neither a ratio above one nor a ratio below one measures successful primary arithmetic. The natural-case A/A paired medians range from 0.977 to 1.012; shuffling and batching reduce but do not remove runtime variation. These observations do not support a claim of a natural-knot acceleration, or a precise cost estimate for an active primary call. Their useful conclusion is that the current corpus and triggering policy do not exercise the new mathematical capability.

All exact ranks and degree counts agree with the original scanner. The Conway and Kinoshita–Terasaka cases are especially useful checks because agreement with simple classical invariants would not itself justify an unknot verdict. The benchmark records the actual planar diagrams and scan orders rather than relying on their names as input specifications.

11.3 Connected algebraic fixtures: successful new splits

Table 2: Bounded algebraic search and complete recursive compression. Trial columns give candidates examined and whether a split was found. Both ratio columns are paired Fitting/primary medians. The starred case raises the variable cap.

r	Mixing	Variables	Old pass trials/split	Primary pass trials/split	Search ratio	Full ratio	Final entries old → primary
3	dense	72	9 / yes	4 / yes	1.161	0.923	13→14
4	dense	128	2 / yes	2 / yes	1.007	0.747	20→20
6	dense	288	8 / yes	2 / yes	1.313	0.964	44→41
8	dense	512	32 / no	2 / yes	2.617	1.015	142→75
10	dense	800	36 / no	2 / yes	2.542	0.966	221→102
12	dense	1152*	40 / no	2 / yes	2.297	0.969	311→156
8	bridge	512	2 / yes	2 / yes	1.010	0.643	40→40

The dense presentations are deterministic conjugates of the two-field family, generated from the recorded seed 2026100809. They test the actual production candidate sequence; they do not sample candidates uniformly as in Theorem 5.2. The explicit bridge presentation from the connectivity proof is included as a negative control.

For $r = 8$ and $r = 10$, the old bounded pass examines 32 and 36 candidates without a split; the primary pass finds a verified split after two candidates. These examples fit the unchanged default limits of 48 objects and 1,024 scalar variables. The $r = 12$ example has 1,152 scalar variables and explicitly raises that allowance. It must not be represented as a successful run under the default cap.

The first-pass time ratios of approximately 2.62, 2.54, and 2.30 compare a completed bounded old search with a successful stronger search. They are *bounded search-pass ratios*, not timings of two algorithms that both finished the same decomposition. The capability improvement is exact: one pass returns a checked split where the other returns no split.

Complete recursive compression is less encouraging as a speed claim. For these three cases, its Fitting/primary ratios are approximately 1.015, 0.966, and 0.969. The primary policy exposes two templates and reduces entries from 142 to 75, 221 to 102, and 311 to 156, while retaining the same number of objects. The recursive code re-solves child commutants and tests further candidates; the initial saving does not persist in the complete pass. These timings do not separately profile the cause of that loss. No material whole-compression wall-clock improvement is established. The bridge example already splits in the old search, and its full compression ratio of 0.643 is a substantial regression for the stronger policy. This supports keeping primary discovery optional while developing better scheduling and reuse.

11.4 Exact probability audits

Table 3: Exact successful-pair counts for the uniform two-field comparison.

r	Fitting successes	Primary successes	All ordered pairs
3	14	44	64
4	30	202	256
5	62	872	1024
6	126	3748	4096

For $r = 3, 4$, every ordered field pair was passed through both matrix algorithms. For $r = 5, 6$, an independent Frobenius-orbit-product calculation examined all field elements and reproduced the complete pair counts. The latter is not an assertion that all 4,096 matrix pairs were individually run. These finite audits corroborate the formula; its validity for every r rests on the proof.

11.5 Geometric expansion avoidance and binary scaling

The expanded comparator explicitly constructs the lifted piece-sheet graph, checks its connected components and orientation double cover, and enumerates boundary cycles. It is an independent correctness oracle and an intentional expansion baseline. It is not an implementation of the compressed Agol–Hass–Thurston machinery, nor an alternative general normal-surface recognizer.

For the two 65,536-sheet annulus examples, compressed preparation takes about 0.086 and 0.123 milliseconds, versus about 292 and 309 milliseconds for the literal expansion. The seven geometry ratios of median control time to median preparation time range from 0.916 to 1.016. The large difference from expansion is robust at this scale, but its precise factor is hardware-

Table 4: Complete geometric preparation versus an independent literal-sheet oracle. Times are median milliseconds per query; this is not a comparison with another compressed algorithm.

Assembly	Sheets	Compressed (ms)	Expanded (ms)
Two Klein bottles	256	0.093	0.453
Two Klein bottles	4,096	0.082	8.952
Two Klein bottles	65,536	0.086	291.679
One torus	65,536	0.123	308.918
Projective-plane/sphere cap	16,384	0.099	88.033
Genus-two base	16,384	0.163	279.716
Eight-patch closed chain	4,096	0.355	125.250

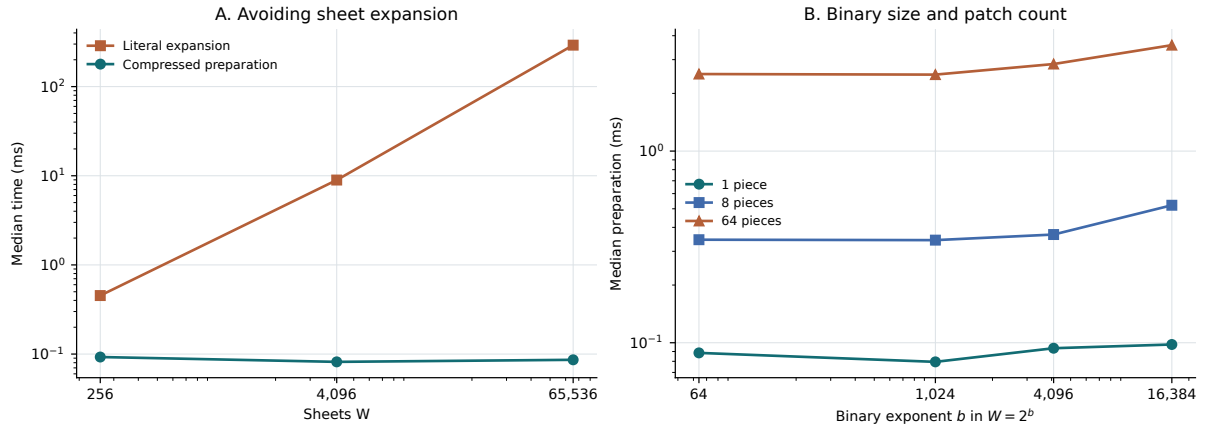


Figure 1: Measured preparation costs from the included raw data. Left: the same annulus family at three explicit sheet counts, with the independent expanded oracle. Right: compressed annulus chains as the binary exponent and number of pieces vary. Connecting segments guide the eye; they are not complexity fits.

and implementation-dependent. The important algorithmic claim is supplied by Theorem 9.1: these operations read binary sheet counts without allocating the sheets.

At $W = 2^{16384}$ and 64 patches, median preparation is 3.571 milliseconds, separate gauge-certificate replay is 1.283 milliseconds, and a prepared 32-mark query is 0.518 milliseconds. The summary serializes to 782,029 bytes and the marked signature to 494,410 bytes. These are compact relative to W , but they are not constant-size outputs. The independent correctness suite also exercises valid gluings at $W = 2^{24000}$; that is an arithmetic and topology test rather than a timed claim about recognition.

11.6 Validation coverage and its limits

The complete final repository suite ran 729 test methods in 102.519 seconds: 726 passed and 3 were skipped because they require the optional native Regina dependency. The pinned baseline had 697 methods and one error in the partial-input worker regression; that error is repaired. The final suite adds 32 methods: twelve primary, five integration, fourteen geometric, and one malformed-byte worker regression. The retained full log records all method names and the independent enumeration totals.

The twelve new primary test methods include exhaustive fixed-space checks for every monic binary modulus of degrees one through six; independent minimal-polynomial checks for all 2×2 binary matrices and seeded 3×3 matrices; repeated factors; nilpotent and identity cases; corrupted certificate rejection; cache separation; interruption; and exact rank comparisons. A

specific attachment test includes two different matching types with recovered quantum shifts zero, one, and two, and checks the complete change of basis. Existing scalar tests include 80 random braid cases. Five integration tests exercise the exact and capped wrappers, recognition and CLI dispatch, option rejection, and exhaustion semantics.

The fourteen new geometry test methods include 1,547 comparisons against literal lifted surfaces and 6,216 ordered-mark comparisons against an independent graph-action oracle. They cover boundary exhaustion, invalid seams, reversed traversal, nonorientable pieces, closed outputs, alternative valid gauge forests, corrupted certificate rejection, component and port provenance, defensive copies, and immutable extension after failure. The $W = 1, 2$ cases are included explicitly because written affine signs are not faithful permutation labels there.

Finite tests do not prove the mathematics or substitute for independent implementation review. Conversely, a proof about a model does not establish that every line of code realizes it. The package supplies the proofs, positive witness checks, small exhaustive comparisons, and a reproducible complete repository suite as complementary evidence. Formal proof-assistant verification remains a separate research task.

12 What would actually imply quasipolynomial recognition?

12.1 A composition bound with all size parameters visible

Let $H(n)$ be the total number of examined global transitions, including failed branches, hierarchy repairs, and rejected certificates. Let $P(n)$ bound the complete bit size of every live representation and operation request: explicit scalar blocks and coefficient data, commutant equations, surface pieces and attachment maps, binary multiplicities, curve or word representations, and generated positive certificates. The maximum ranges over every intermediate representation actually materialized, including temporary expansions and the state before recompression. Let all local work for one transition, including construction, validation, lookup, output, and any required subroutine, be bounded by $P(n)^c$ for a constant c .

Proposition 12.1 (Conditional composition bound). *Under these hypotheses, a correct complete recognition algorithm takes*

$$T(n) \leq \text{poly}(n) + O(H(n)P(n)^c).$$

If $\log H(n)$ and $\log P(n)$ are both $O((\log(n+2))^2)$, this is $n^{O(\log n)}$. More generally, quasipolynomial bounds on both H and P give a quasipolynomial bound on T .

Proof. Charge each local operation, including its unsuccessful attempts, to the transition performing it. By hypothesis the total charge per transition is at most a constant times P^c . Summing gives the first expression. Taking logarithms proves the remaining statements. Any subroutine lacking the asserted bound is a missing hypothesis, not a term that may be omitted. $\square$

This elementary proposition is useful because it exposes the actual gap. Theorems 9.1 and the polynomial-projector cost bound justify some local charges. They do not bound the live scalar dimensions, prove that geometric requests have the required affine form, or bound the number of transitions. A local algorithm polynomial in an exponentially large explicit matrix is still exponential in the original crossing count. An algorithm polynomial in $\log W$ is more favorable when a large multiplicity has remained genuinely binary throughout its construction.

A bounded heuristic that eventually falls back to complete Khovanov scanning remains a valid recognizer when sufficient resources are allowed, using the established unknot-detection theorem

[5]. For completeness, the coefficient bridge is as follows. For a knot K ,

$$\dim_{\mathbb{F}_2} Kh(K; \mathbb{F}_2) = 2 \dim_{\mathbb{F}_2} \widetilde{Kh}(K; \mathbb{F}_2),$$

by the characteristic-two splitting [6, Corollary 3.2.C]. Universal coefficients give

$$\dim_{\mathbb{F}_2} \widetilde{Kh}(K; \mathbb{F}_2) \geq \dim_{\mathbb{Q}} \widetilde{Kh}(K; \mathbb{Q}),$$

and the latter is greater than one for every nontrivial knot by the Kronheimer–Mrowka theorem. The unknot has reduced rank one, so unreduced characteristic-two rank two is equivalent to being the unknot. That logical completeness does not improve its general exponential bound. With fixed resource limits it is a sound partial recognizer whose UNKNOWN outputs record precisely that distinction.

12.2 Hierarchy depth, pattern complexity, and the exponent

Suppose a hierarchy procedure has an iteration bound

$$H \leq L(g+1)^L, \tag{16}$$

as in the cited hierarchy framework, and suppose every iteration satisfies the local cost hypothesis. Then

$$\log T \leq O(\log n) + O(\log L) + L \log(g+1) + c \log P. \tag{17}$$

The three terms involving L, g, P have distinct mathematical origins. Improving a normal-cut implementation does not automatically improve the repair term $L \log(g+1)$.

Hierarchy length L	Pattern bound g	Representation P	Resulting bound
$O(\log n)$	$n^{O(1)}$	$n^{O(1)}$	$2^{O((\log n)^2)}$
$O((\log n)^2)$	$n^{O(1)}$	$n^{O(1)}$	$2^{O((\log n)^3)}$
$O((\log n)^2)$	$O(1)$	$n^{O(1)}$	$2^{O((\log n)^2)}$

Each row is conditional on the complete per-iteration accounting. The last row points to an alternative to reducing hierarchy depth: sufficiently strong control of the pattern choices could achieve the sharper exponent at the same depth. No such uniform pattern bound is proved here.

The geometric kernel helps with a specific part of P and its local cost. A parallelity region represented as checked dihedral cover pieces can be sewn and queried without replacing a binary degree by that many explicit components. To use this fact in (17), one must still extract the pieces and seams from the embedded normal-cut data and prove closure of the representation under every subsequent operation. The compressed-cut theorem in the 2026 preprint is a significant relevant input; it does not by itself discharge all of Section 9’s operation costs [8].

12.3 A sharper sufficient condition for variable lexicographic bounds

A single worst-case g can hide useful variation between hierarchy levels. Suppose the progress measure is a fixed-length vector with coordinate bounds $0 \leq x_i \leq G_i$ that apply uniformly throughout every examined history, and each macrostep strictly decreases its lexicographic value. Coordinates are padded consistently so that changing hierarchy length does not evade the order. Then the number of possible values is at most

$$\prod_{i=1}^L (G_i + 1).$$

If at most $F(n)$ charged operations occur between consecutive decreases, including initialization and termination, then $H(n) \leq F(n) \prod_i (G_i + 1)$. The relevant exponent budget, in addition to $\log F(n)$, is therefore

$$\sum_{i=1}^L \log(G_i + 1), \quad (18)$$

rather than just the largest coordinate or the length of the written vector.

For example, proving (18) to be $O((\log n)^2)$ would suffice for the sharper target when P and the per-value factor satisfy the same logarithmic bound. A small binary encoding of a decreasing integer does not alone imply few transitions: a b -bit counter may still take 2^b distinct values. Equally, decreasing one coordinate while allowing lower coordinates to reset is sound only when their complete ranges remain bounded. These are explicit obligations for a proposed amortized hierarchy proof.

12.4 Algebraic state sharing has a corresponding obstruction

The same accounting applies to compressed Khovanov states. Splitting a component may expose repeated templates and reduce later operations, but it may also leave object counts unchanged, as the measured examples do. A theorem would need a bound on the number and size of distinct live templates, the bit lengths of their multiplicities and grading data, and the cost of updating all attachments as crossings are added. Neither semisimplicity of a chosen commutative subalgebra nor a successful split at one stage bounds those quantities at the next stage.

The two-field construction also warns against treating candidate behavior as universal. Uniform random field values give the favorable probability in Theorem 5.2; repeated isomorphic primary factors can make every polynomial in one candidate fail to distinguish certain summands. A general representation theorem or a genuinely broader endomorphism algorithm is needed before assigning a global asymptotic benefit to this optional kernel.

12.5 A concrete next optimization: reuse the corner algebras

The complete-compression measurements suggest a specific improvement that has a short exact justification. Let $R = \text{End}_{\text{scalar}}^0(C)$ be the complete typed scalar chain-endomorphism algebra, and let $1 = e_1 + \cdots + e_s$ be pairwise orthogonal typed scalar idempotents giving a verified decomposition $C = \bigoplus_i C_i$ with the inherited scalar types.

Proposition 12.2 (Transported corner spaces). *There are natural identifications*

$$\text{End}_{\text{scalar}}^0(C_i) \cong e_i R e_i, \quad R = \bigoplus_{i,j} e_i R e_j \quad \text{as vector spaces.}$$

In particular, a complete vector-space basis $R_1, \dots, R_a$ of R provides spanning sets $\{e_i R_j e_i : 1 \leq j \leq a\}$ for all child commutants, and

$$\sum_i \dim_{\mathbb{F}_2} \text{End}_{\text{scalar}}^0(C_i) \leq \dim_{\mathbb{F}_2} R.$$

Proof. Write the inclusion and projection of C_i as u_i, v_i , so that $u_i v_i = e_i$ and $v_i u_i = 1$. A typed scalar endomorphism a of C_i extends to $u_i a v_i \in R$ supported on that summand. Conversely, restricting $e_i r e_i$ gives $v_i r u_i$. These maps are inverse and preserve multiplication. Every $r \in R$ equals $\sum_{i,j} e_i r e_j$; multiplying a linear relation on the left and right by chosen idempotents proves that this sum of subspaces is direct. The dimension inequality follows by retaining only diagonal terms. $\square$

Thus a static decomposition pass need not solve the original differential commutation equations afresh for every child. It can conjugate and restrict the already computed basis, discard linear dependencies, and pass the resulting corner basis down the decomposition tree. This is an application of a standard corner-algebra identity, not a new structural theorem about finite-dimensional algebras. Its value here is that it isolates a reusable object that the current implementation discards.

The inequality gives an amortized *dimension* budget at every frontier of a static refinement tree. Every nonzero child has a nonzero identity, so a proper refinement tree has at most a leaves; it also has at most D leaves by total scalar dimension. If each internal node has at least two nonzero children, there are at most $\min(a, D) - 1$ internal refinements. A simultaneous conjugation of a explicit D -dimensional basis matrices by the splitting basis and its inverse costs $O(aD^3)$ binary operations; all child matrices are then restrictions of the conjugated matrices. At any depth the child commutant dimensions sum to at most a , so total basis-conjugation work over the accepted static tree is bounded by $O(aD^3 \min(a, D))$, hence $O(aD^4)$. Candidate search, dependency elimination, and differential-attachment transport are additional costs. Alternative or backtracking trees are also outside this particular bound. Simultaneous primary refinement can reduce the number of separate transports. This optimization is proposed and proved at the interface level, but is not implemented or included in the reported speed figures.

Adding a crossing or changing the differential invalidates the simple static argument. Reusing a corner basis through those mutations requires a new exact update rule or a fresh solve. The first implementation should therefore target one existing `_compress` call, where the benchmark exhibits repeated child solves as an explicit operation to profile.

13 Proposed research questions and work program

The following questions are ordered by their relation to the demonstrated kernels and to the eventual recognition theorem. Each has a specific mathematical or experimental deliverable. They should be treated as open work items, not as assumptions already validated by this article.

13.1 Near-term algebraic questions

1. Can corner reuse turn successful splitting into a full-pass gain? Implement Proposition 12.2 inside one static compression pass. Retain the original complete commutant basis, transport it through every accepted basis change, and derive child bases by corner restriction and independent elimination. Compare identical decompositions, complete wall clock, transported matrix volume, and peak memory. A successful result would preserve the current witness checks while reducing complete-pass cost on $r = 8, 10$ without a comparable regression on the bridge and natural controls. The proposed optimization should first be tested with the same canonical commutant bases or a replay of the same candidates in both arms. Merely fixing the policy does not control candidate changes caused by a different input basis.

2. Which trigger policy makes primary discovery worthwhile? A primary candidate may be cheap to reject or expensive to analyze, and the old basis order can already expose easy idempotents. Develop a cost model using block sizes, estimated minimal-polynomial degree, remaining split allowance, and the measured cost of a commutant solve. Compare immediate fallback, delayed fallback after the ordinary candidate pass, and shared power tables. This is an algorithm-selection problem, not a license to omit unfavorable cases. Record primary invocations, positive splits, failed-call time, child-solve time, and total recognition time.

3. Can all polynomial summands be extracted in one certified pass? Use a basis of the Berlekamp algebra to refine idempotents by $e \mapsto ep$ and $e \mapsto e(1 + p)$, removing zero pieces. Berlekamp’s classical argument identifies all primary factors of the chosen cyclic algebra. The new task is to obtain one combined typed basis change, retain grading and all attachments, and compare its matrix-transport cost with recursive binary splitting. This remains incomplete for the full noncommutative commutant, so the output contract must say exactly which family of projectors was exhausted.

4. Can the comparison family occur in genuine knot computations? Determine whether the connected $xI + yA$ family, or a comparable family with the same scalar endomorphism obstruction, occurs as a reduced prefix of a classical knot diagram under the repository’s exact scanner. A proof of realizability would connect the candidate separation to the target domain. A proof of impossibility would be equally informative: it could reveal relations satisfied by reachable prefixes that generic category objects miss. Finite searches should retain the original diagram, crossing order, all basis changes, and a replayable provenance chain. Merely finding an isomorphic matrix after deleting external attachments is insufficient.

5. What can replace polynomial-in-one-operator search? Repeated primary types are invisible to one cyclic subalgebra. Study the semisimple quotient and radical of the full typed scalar commutant, or several interacting candidates, with explicit field and bit costs. A useful target is a deterministic decomposition routine whose accepted idempotents have the same small positive certificates as the present backend. One must not linearize $Q \mapsto Q^2 + Q$ on a noncommutative algebra: the cross terms make that map nonlinear. Every use of Frobenius linear algebra must identify its commutative domain.

6. Is there a size theorem for reachable shared states? Seek structural bounds on distinct templates, template dimensions, multiplicity bit lengths, and attachment complexity after each crossing. A bound on boundary width alone does not bound arbitrary multiplicity spaces unless the representation theorem explicitly says so. Start with restricted families such as controlled braid width or bounded separator profiles and state the dependence on each parameter. The decisive outcome would be a bound strong enough for Proposition 12.1, not only smaller constants on a fixed corpus.

13.2 Geometric representation and attachment questions

7. Can real normal cuts produce checked seam presentations? Implement an extractor from a triangulation or uniform-type handle structure and compressed normal coordinates to the piece/seam input of Section 6. The extractor must derive peripheral words, local orientations, fibre coordinates, and seam maps from the actual embedded construction. Its certificate must connect the abstract cover to the cut manifold’s attaching data. First establish a restricted theorem for a recognizable dihedral parallelity configuration; reject inputs outside that class. Guessing an affine action from component counts would lose the information highlighted by Example 9.2.

8. How far can the full-boundary promise be relaxed? Normal-surface orbit systems can involve partial intervals and multiple coordinate ranges. Investigate whether such data admit a bounded number of signed-affine charts, or require the more general interval-pairing machinery. A precise extension should state its allowed breakpoints, binary coordinate lengths, chart overlap maps, and composition rule. Measure complexity before normalizing or refining

all charts. Prove a bound on newly introduced breakpoints; otherwise avoiding sheet expansion can simply move the exponential growth into the number of intervals.

9. Can seam insertion be maintained incrementally? The present immutable extension reconstructs its index. A dynamic version could maintain affine potentials on a disjoint-set forest, with new cycle generators updating gcd and orientation data. The difficulty is preserving all old piece coordinates, component labels, fixed-residue exceptions, boundary availability, and ordered-mark queries when components merge. A useful theorem would give an amortized bit bound including large-integer arithmetic and the cost of emitted updates. Returning a full changed summary after each insertion still costs its full output size.

10. Which embedded attachment queries are sufficient for hierarchy repair? Point and boundary-lift labels are only part of a 3-manifold boundary pattern. Identify the exact queries used by the hierarchy’s repair steps: for example, curve isotopy, intersection, relative position, product-region attachment, and the distinction between compressible and essential patterns. Relate them to compressed surface-curve algorithms such as Lackenby’s normal-curve procedures [9]. The main issue is constructing their required inputs with controlled length while preserving embedding provenance. An abstract homeomorphism type and a list of marked points do not by themselves solve these questions.

13.3 Global complexity and research infrastructure

11. Can hierarchy progress satisfy a total logarithmic budget? For an explicit hierarchy algorithm, identify the actual progress vector, its coordinate ranges, and every transition that changes it. Prove either a bound on $L \log(g + 1)$ or the more precise sum in (18). Investigate whether geometric simplification reduces the number of genuinely variable pattern choices at most levels. A logarithmic depth bound, a constant pattern bound, and a weighted coordinate bound are different routes with different proof obligations. Adopt the exponent ledger (17) before optimizing an individual subroutine, so each improvement has a visible global role.

12. Which hard instances exercise the new operations? Build a versioned corpus that separates filter exits, ordinary scalar splits, active primary failures, active primary successes, large affine cover degrees, and normal-cut extraction failures. Include hard unknot presentations and knotted controls with weak classical invariants, and retain exact input diagrams and generation histories. Transforming one knot diagram many times should not be confused with sampling independent knot types. Report per-family results and unfavorable tails, rather than a single aggregate speed factor. Search-generated witnesses should be frozen before comparing new trigger policies.

13. What should a small independently trusted checker verify? Specify a certificate language for scalar decompositions and geometric attachments. For algebra, the positive projector and conjugation checks already provide a starting point; a complete commutant-basis certificate would additionally need rank/completeness evidence. For geometry, extend the gauge certificate with the normal-cut extraction witness rather than trusting a serialized genus. Formalize the small algebraic and covering lemmas in the proof system used by the repository, with exact treatment of binary integers, $W = 1, 2$, orientation characters, and interruption states. The current Python tests and prose proofs are not machine-checked formal proofs.

14. Can a portfolio retain a proved worst-case bound? A practical recognizer may combine cheap filters, Khovanov methods, and a geometric hierarchy. To inherit a future quasipolynomial theorem, give the proved branch a fair, quantitatively bounded share of resources and charge all speculative work. An unbounded heuristic performed first can destroy a worst-case guarantee even when a good fallback exists. Conversely, bounded optional algebraic work can coexist with a proved geometric bound. The scheduling theorem must include preprocessing, subprocess startup, certificate checking, and the cost of converting between representations.

13.4 Suggested order of work

The most concrete implementation step is corner reuse, followed by a controlled comparison of primary trigger policies. The most consequential geometric step is a certified extractor on a restricted real normal-cut family. Those projects can proceed independently and share a common witness and benchmark discipline. A global asymptotic claim should wait for explicit representation-closure and progress bounds; the local proofs in this article can then be inserted as established subroutines rather than rediscovered during the global argument.

14 Reproduction and repository integration

The accompanying archive is intended to be reviewed and integrated at the pinned baseline. It contains the article source and PDF, generated tables and figure, the complete relevant source snapshot, the required historical test oracles, raw numerical evidence, validation logs, a baseline patch, and file hashes. Historical benchmark-result directories unrelated to this continuation are omitted from the snapshots; the test fixtures and source files needed by the maintained suite are retained.

The source snapshot is under `source/Topology/UnknotRecognition/`. The `fast` directory remains a normal standalone Python package. The historical `reports/24/reference`, `reports/26/detshadow`, and `reports/28/src` modules are retained because existing tests import them. Their presence is a reproduction dependency, not a claim that their older results are new contributions.

```
# Run from the extracted archive root.
python scripts/reproduce.py --verify
python scripts/reproduce.py --tests
python scripts/make_figures.py
cd article
pdflatex -interaction=nonstopmode -halt-on-error unknot_projectors_and_seams.tex
pdflatex -interaction=nonstopmode -halt-on-error unknot_projectors_and_seams.tex
```

The verification command checks archive hashes and the recorded measured source hashes. The test command runs the complete maintained suite from the packaged source. Benchmarks are opt-in because their runtime is substantially longer than a single example:

```
python scripts/reproduce.py --benchmarks
# Individual operations, from source/.../fast:
python -m fastunknot recognize examples/trefoil.json --backend primary
python -m fastunknot khovanov examples/trefoil.json --primary
python -m fastunknot.surface_gluings gluing_research/examples/torus_twenty.json
```

The fresh benchmark files use a separate output directory and do not replace the retained measurements. Wall-clock samples will vary by machine. The geometry driver's metadata

collection has a small portability change after the measured run: outside a Git checkout it now records a null commit instead of failing. The exact measured driver is retained under `evidence/measured_sources`; the arithmetic kernels and timed operations are unchanged. This distinction is explicit in the source manifest.

The integration patch contains both new kernels, public dispatch, tests, research fixtures and drivers, and the carried-forward worker repair with the additional malformed-byte regression. Apply it only after checking the baseline or resolving later repository changes. The package’s baseline snapshot permits a local patch-application check without a network fetch. The supplied article is a research continuation with proved restricted results and documented open problems; it is not a completed general quasipolynomial recognition proof.

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